

M. R. Tanush Pai

AI/ML Engineer · Generative AI & LLM Systems · Agentic AI · Python Backend Engineering
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PROFESSIONAL SUMMARY

AI/ML Engineer and incoming graduate with hands-on experience engineering production-grade Generative AI applications and agentic workflows. Demonstrated capability in Python backend development, designing scalable FastAPI architectures, and deploying custom LLM platforms that automate complex enterprise data workflows. Adept at bridging the gap between advanced machine learning algorithms and robust full-stack software delivery.

TECHNICAL SKILLS

Languages: Python, SQL, R

AI/ML & GenAI: LangChain, LangGraph, Scikit-learn, Hugging Face, OpenAI API, RAG, Agentic Workflows, Vector Search, Prompt Engineering, Ollama

Data & Databases: Pandas, NumPy, PostgreSQL, MongoDB, Redis, FAISS

Backend & Tools: FastAPI, Prisma, Vercel, Git, GitHub, Bitbucket, Jira

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern | Sam Corporate

Kochi, India

Feb 2026 – Present

- Architected an agentic AI analytics chatbot with an LLM-based planner that dynamically generates MongoDB aggregation pipelines across 25+ collections, enabling non-technical staff to query data using natural language with GDPR-aligned privacy controls.
- Built an AI-powered QMS document generation platform adopted by 12+ employees, automating content creation across 10+ templates via LLMs and reducing document turnaround from hours to minutes.
- Extended an ESG analytics platform with 8+ new modules — including AI-driven report generation, NL2SQL, peer benchmarking, and interactive chart generation — using PostgreSQL, OpenAI, and Perplexity APIs.
- Resolved a production LLM reliability issue by optimizing prompt size and API workflows, eliminating HTTP 429 rate-limit errors and stabilizing application performance.
- Developed an automated MongoDB data validation tool that cross-checks records across multiple databases and exports Excel reports, replacing manual spreadsheet workflows with a single-click process.

Freelance Full-Stack Developer | Chlonestone

Dubai, UAE (Remote)

May 2026 – Jun 2026

- Developed a production-grade full-stack real estate platform and CRM for a Dubai-based client using Next.js, TypeScript, Prisma, PostgreSQL, Tailwind CSS, and Supabase.
- Built property listing, developer discovery, community exploration, lead capture, inquiry management, and advanced search modules for customer engagement.
- Engineered an internal CRM to manage agents, leads, subscribers, follow-ups, and customer engagement workflows, integrating automated email notifications and Supabase Storage for secure media management.
- Deployed the application on Vercel and optimized the architecture for scalable production deployment and sales operations.

KEY PROJECTS

AVANA – Autonomous Visual Analytics & Normative Agents | Python, LangGraph, LangChain, Ollama, FLAML

- Architected a 3-agent LangGraph pipeline (*Data Engineer* → *Data Analyst* → *Data Scientist*) that autonomously transforms raw CSV uploads into cleaned datasets, EDA reports, and trained ML models with zero manual intervention.
- Implemented a Master Orchestrator with a shared Dataset Contract (target column, problem type, domain context) coordinating agents across a 13-stage preprocessing pipeline, 50+ automated visualizations, and FLAML AutoML training.
- Integrated an AI Consultant chat interface (Ollama / Llama 3.1) with role-aware personas for contextual interaction with pipeline outputs, visualizations, and model insights.

AcademicXchange | HTML, CSS, JavaScript, Python, Hugging Face

- Built a responsive educational platform for KTU students providing centralized access to notes, syllabi, and past question papers.

- Implemented an AI Question Paper Generator and a PDF Summarizer chatbot using Hugging Face models to auto-generate papers, summarize documents, and answer student queries interactively.

EDUCATION

SCMS School of Engineering and Technology

B.Tech in Computer Science (Data Science)

Kerala, India

Oct 2022 – Apr 2026

CERTIFICATIONS

- Multi AI Agent Systems with CrewAI – DeepLearning.AI, Coursera (2025)
- Fundamentals of Building AI Agents – IBM, Coursera (2025)
- Data Analysis with Python and Pandas – Infosys Springboard (2025)
- Python for Data Science – Infosys Springboard (2025)